

$$11.9 \text{ (a) Since } \chi'^{\alpha} = (g^{\alpha\beta} + \epsilon^{\alpha\beta}) \chi_{\beta} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})(g_{\beta\gamma} + \epsilon'_{\beta\gamma}) \chi'^{\gamma} \\ = (f^{\alpha}_{\gamma} + \epsilon^{\alpha}_{\gamma} + \epsilon'_{\gamma} + 0) \chi'^{\gamma} = \chi'^{\alpha} + (\epsilon^{\alpha\beta} + \epsilon'^{\beta\alpha}) \chi'_{\beta}.$$

$$\text{then } \epsilon^{\alpha\beta} + \epsilon'^{\beta\alpha} = 0, \text{ or } \epsilon'^{\beta\alpha} = -\epsilon^{\alpha\beta}$$

(b) By the preservation of norm,

$$\chi'^{\alpha} \chi'_{\alpha} = g_{\alpha\beta} \chi'^{\alpha} \chi'^{\beta} = g_{\alpha\beta} (g^{\alpha\gamma} + \epsilon^{\alpha\gamma})(g^{\beta\delta} + \epsilon'^{\beta\delta}) \chi_{\gamma} \chi_{\delta} \\ = (\delta^{\alpha}_{\beta} + \epsilon^{\alpha}_{\beta})(g^{\beta\gamma} + \epsilon'^{\gamma\beta}) \chi_{\gamma} \chi_{\delta} = (g^{\mu\nu} + \epsilon^{\nu\mu} + \epsilon^{\mu\nu} + 0(\epsilon^2)) \chi_{\mu} \chi_{\nu} \\ = \chi^{\alpha} \chi_{\alpha} + (\epsilon^{\alpha\beta} + \epsilon^{\beta\alpha}) \chi_{\alpha} \chi_{\beta}$$

$$\text{then } (\epsilon^{\alpha\beta} + \epsilon^{\beta\alpha}) \chi_{\alpha} \chi_{\beta} = 0, \text{ or } \epsilon^{\alpha\beta} = -\epsilon^{\beta\alpha}$$

(c) By (b), $\epsilon^{\alpha\beta}$ is antisymmetric, while L is also antisymmetric. By choosing the upper diagonal matrix elements, we can make $\epsilon^{\alpha\beta}$ to be identical to L .